

1. Three particles of masses $2m$, $3m$ and $4m$ are placed at the points with coordinates $(-2, 5)$, $(2, -3)$ and $(3k, k)$ respectively, where k is a constant.
The centre of mass of the three particles is at the point $(\bar{x}, \bar{y})$.

(a) Show that $\bar{x} = \frac{2 + 12k}{9}$ **(2)**

The centre of mass of the three particles lies at a point on the straight line with equation $x + 2y = 3$

(b) Find the value of k . **(4)**

2. A car of mass 900 kg is moving down a straight road which is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{12}$

The engine of the car is working at a constant rate of 15 kW.

The resistance to the motion of the car is modelled as a constant force of magnitude 400 N.

Find the acceleration of the car at the instant when it is moving at 16 m s^{-1}

(5)

3. A particle P of mass 0.2 kg is moving with velocity $(4\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$.
The particle receives an impulse $\lambda(\mathbf{i} + \mathbf{j}) \text{ N s}$, where λ is a constant.
Immediately after receiving the impulse, the speed of P is 7 m s^{-1} .
Find the possible values of λ .

(6)

4. At time t seconds ($0 \leq t < 5$), a particle P has velocity $\mathbf{v} \text{ m s}^{-1}$, where

$$\mathbf{v} = (\sqrt{5-t})\mathbf{i} + (t^2 + 2t - 3)\mathbf{j}$$

When $t = \lambda$, particle P is moving in a direction parallel to the vector $\mathbf{i}$.

- (a) Find the acceleration of P when $t = \lambda$

(5)

The position vector of P is measured relative to the fixed point O

When $t = 1$, the position vector of P is $(-2\mathbf{i} + \mathbf{j})\text{m}$.

Given that $1 \leq T < 5$

- (b) find, in terms of T , the position vector of P when $t = T$

(5)

5.

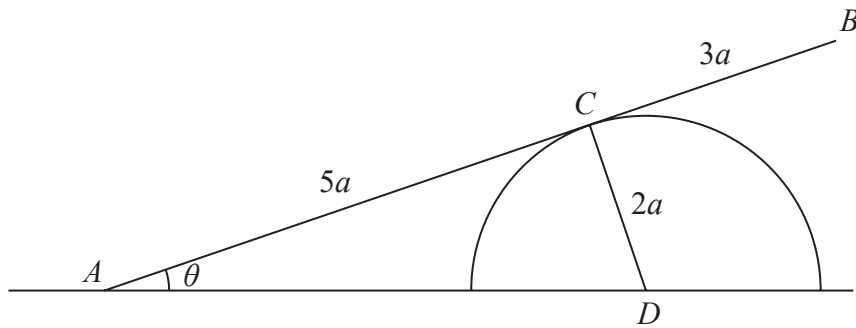


Figure 1

A uniform rod AB has length $8a$ and weight W .

The end A of the rod is freely hinged to horizontal ground.

The rod rests in equilibrium against a block which is also fixed to the ground.

The block is modelled as a smooth solid hemisphere with radius $2a$ and centre D .

The point of contact between the rod and the block is C , where $AC = 5a$

The rod is at an angle θ to the ground, as shown in Figure 1.

Points A , B , C and D all lie in the same vertical plane.

(a) Show that $AD = \sqrt{29}a$ (1)

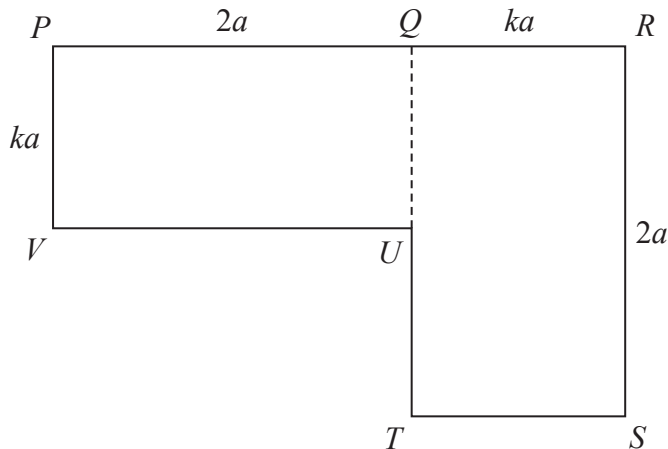
(b) Show that the magnitude of the normal reaction at C between the rod and the block is $\frac{4}{\sqrt{29}}W$ (3)

The resultant force acting on the rod at A has magnitude kW and acts at an angle α to the ground.

(c) Find (8)

- (i) the exact value of k
- (ii) the exact value of $\tan \alpha$

6.

**Figure 2**

The uniform lamina $PQRSTUV$ shown in Figure 2 is formed from two identical rectangles, $PQUV$ and $QRSTU$.

The rectangles have sides $PQ = RS = 2a$ and $PV = QR = ka$.

- (a) Show that the centre of mass of the lamina is $\left(\frac{6+k}{4}\right)a$ from PV (3)

The lamina is freely suspended from P and hangs in equilibrium with PR at an angle of α to the downward vertical.

Given that $\tan \alpha = \frac{7}{15}$

- (b) find the value of k .

(6)

7. Particle A has mass m and particle B has mass $2m$.

The particles are moving in the same direction along the same straight line on a smooth horizontal surface.

Particle A collides directly with particle B .

Immediately before the collision, the speed of A is $3u$ and the speed of B is u .

The coefficient of restitution between A and B is e .

- (a) (i) Show that the speed of B immediately after the collision is $\frac{5 + 2e}{3}u$

- (ii) Find the speed of A immediately after the collision.

(7)

After the collision, B hits a smooth fixed vertical wall that is perpendicular to the direction of motion of B .

The coefficient of restitution between B and the wall is $\frac{1}{3}$

Particle B rebounds and there is a second collision between A and B .

The first collision between A and B occurs at a distance d from the wall.

The time between the two collisions is T .

Given that $e = \frac{1}{2}$

- (b) find T in terms of d and u .

(6)

8. [In this question, the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a vertical plane, with $\mathbf{i}$ being horizontal and $\mathbf{j}$ being vertically upwards.]

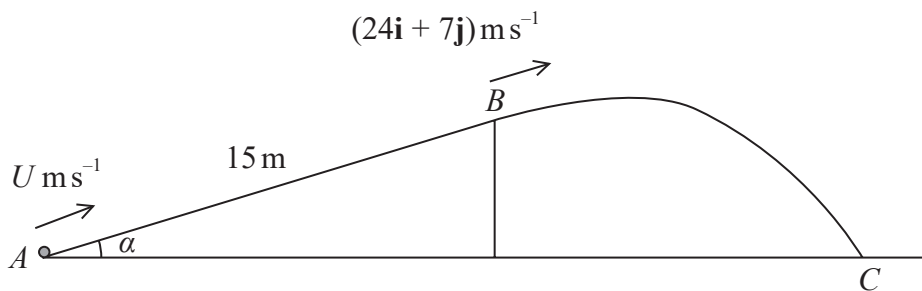


Figure 3

A rough ramp is fixed to horizontal ground.

The ramp is inclined to the ground at an angle α , where $\tan \alpha = \frac{7}{24}$

The point A is at the bottom of the ramp and the point B is at the top of the ramp.
The line AB is a line of greatest slope of the ramp and $AB = 15 \text{ m}$, as shown in Figure 3.

A particle P of mass 0.3 kg is projected with speed $U \text{ m s}^{-1}$ from A directly towards B .
At the instant P reaches the point B , the velocity of P is $(24\mathbf{i} + 7\mathbf{j}) \text{ m s}^{-1}$
The particle leaves the ramp at B , and moves freely under gravity until it hits the horizontal ground at the point C .

The coefficient of friction between P and the ramp is $\frac{1}{5}$

(a) Find the work done against friction as P moves from A to B . (3)

(b) Use the work-energy principle to find the value of U . (4)

(c) Find the time taken by P to move from B to C . (3)

At the instant immediately before P hits the ground at C , the particle is moving downwards at θ° to the horizontal.

(d) Find the value of θ (4)